

Application shell

operational program boundary

config file, generated command line, logging, daemonization, quiet mode
diagnostic evidence: effective configuration, output mode, startup state



Instance scope

configured server/client role visible at runtime

enabled/disabled state, listen/connect activation, retry scheduling
diagnostic evidence: role lifecycle, endpoint selection, activation result



Connection scope

one concrete peer relationship

connection name, local/remote addresses, queued/sent/read/processed data
diagnostic evidence: peer episode, counters, duration, disconnect summary



Context scope

protocol behavior attached to one connection

protocol state, accepted/rejected frames, application-level events
diagnostic evidence: protocol meaning, state transitions, close decisions

Logging forms

LOG(level)
ordinary lifecycle
and severity

PLOG(level)
lifecycle plus captured
system-error context

VLOG(n)
optional diagnostic
depth